

What is left before this can take real money

The commerce build is substantially complete and has never been deployed. Everything below is either a key, a piece of infrastructure, a supplier email, or a legal appointment — and four of those are hard gates that no amount of engineering removes.

Stack Medusa 2.18 + Next.js 16 + Postgres 17 Catalog 4,323 products / 29,002 variants Tests 537 passing
Deployed nowhere

HARD BLOCKERS 9 5 API keys, R2, hosting, and the payment path's last step	CANNOT BE CODED 4 IOSS, GDPR Art. 27, GPSR person, licensing. Start today	ONE SUPPLIER EMAIL 5 gaps Fibre, origin, HS codes, measurements, print format	BUILT AND TESTED Most of it Checkout, returns, accounts, tax, fulfilment, personalisation
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READ THIS FIRST

The application currently cannot boot in production, and filling in the configuration it asks for will not fix it. Cloudflare R2 is marked `criticalInProduction`, so `assertConfigured()` throws at start-up while its keys are empty — but no code anywhere reads them, and `MEDIA_BACKEND=r2` exists only in two comments. Images are served from Postgres and work. This is a five-minute decision (build the adapter, or set the flag to `false`) standing in front of every other deployment task.

01

Front end

Twenty routes, all rendering, 0 mechanical accessibility issues across 25 audited pages, WCAG AA contrast verified by script in CI. What remains is small and mostly not code.

ITEM	WHY IT MATTERS	WHEN
Product imagery	67% of products have exactly one photograph. §13.6 accepted that deliberately and named the cost: fit and appearance surprises drive apparel returns, and under a final-sale policy the customer absorbs that rather than us. Photograph the top sellers once there is order data.	AFTER LAUNCH
Screen-reader pass	The mechanical audit covers 25 pages and now checks landmark naming, radio grouping, live regions and the skip link. Nobody has listened to VoiceOver or NVDA work the buy box and checkout, and nobody has done a keyboard-only traversal.	BEFORE LAUNCH
Size measurements	<code>MEASUREMENTS</code> is empty, so <code>/size-guide</code> says so rather than printing a generic chart. Under final sale a wrong size is the customer's loss, which makes this the highest-value line in the supplier email.	BEFORE LAUNCH
Personalisation teaser	The homepage block is still a placeholder. The preview itself is live on every product page; a teaser that cannot lead to a finished order is not worth the space until the print-file format exists.	AFTER LAUNCH
Search quality	Free-text search is <code>ILIKE '%...%'</code> , which cannot use an index. Fine at 4,300 products and a real problem at 40,000 or under load. Typesense is registered and has no implementation behind it.	AFTER LAUNCH

ITEM	WHY IT MATTERS	WHEN
Favicon and brand assets	The default Next.js favicon is still in place. <code>public/</code> was emptied of the starter SVGs; nothing replaced them.	BEFORE LAUNCH

02

Back end

Keys — five of nine integrations block a production boot

Every module behind these is written and tested. The rule in `src/integrations.ts` is that a missing key degrades visibly outside production and refuses to start inside it, so each row below is a hard stop rather than a degraded feature.

INTEGRATION	STATE	WHEN
Stripe	Payment path proven 7 of 8 steps; the last needs a test key. With STRIPE_WEBHOOK_SECRET empty the webhook endpoint returns HTTP 200 to unsigned payloads — measured, not assumed. Must be set before this is reachable from the internet.	BLOCKER
Resend	9 templates, all wired. Nothing sends without the key, and order confirmations do not exist until it is set.	BLOCKER
Stripe Tax	Module written and tested, including that it sends cents rather than dollars. Distinguishes “no tax owed” from “tax not switched on”. One HTTP call to Stripe is untested.	BLOCKER
Shippo	Rating is already live off our own rate card. Only labels, tracking and customs documents need the key.	BLOCKER
Cloudflare R2	Marked critical and not implemented. See the note on page 1. Either build the adapter or stop marking it critical.	BLOCKER
Sentry	Custom reporter, no SDK. Fingerprinting and scrubbing already work without a DSN; nothing is being watched until it is set.	BEFORE LAUNCH
PostHog	Consent-gated loader written. No funnel data until the key is set.	BEFORE LAUNCH
Typesense	Registered with no implementation . A key would do nothing.	AFTER LAUNCH
Omnisend	Registered with no implementation . Subscribers are being collected so the list exists when one is chosen.	AFTER LAUNCH

Correctness and scale

ITEM	DETAIL	WHEN
Event bus and locking	Medusa says it at every boot: <i>“Local Event Bus installed. This is not recommended for production”</i> , and locking is in-memory. Jobs run in-process, so a restart loses queued work — including order confirmation emails. Redis is already running and is the fix.	BLOCKER
In-process caches	Four endpoints cache in module memory: facets, sitemap, store reviews, collections. Two app instances will disagree with each other, and an import is invisible until a TTL expires. Correct on one instance, wrong on two.	BEFORE LAUNCH
Inventory	<code>manage_inventory</code> is false on all 29,002 variants. That is a decision for a sourcing model, not an oversight — but it means nothing can ever be out of stock, including things the supplier cannot get.	AFTER LAUNCH

ITEM	DETAIL	WHEN
Print-file generator	The only genuinely blocked piece of engineering. Needs the supplier's required format — vector or raster, spot colours, template. The on-screen preview is independent and finished. Two personalisations are already queued and nothing can print them.	BLOCKER
Invoice PDFs	Required for EU B2C sales only. A US-first launch does not need them.	AFTER LAUNCH
Refund and label execution	The returns queue records decisions and deliberately does not move money or buy labels: both are irreversible external calls and Shippo's transaction endpoint is not idempotent. Somebody has to do that step in Orders, or it needs building with an idempotency key.	BEFORE LAUNCH

03

Infrastructure — none of this exists yet

This is the largest remaining block of work and the least interesting. Nothing has ever been deployed; `docker-compose.yml` is production-*shaped* (built images, health checks, no bind mounts) and has never been run against a real environment. `.env.deploy` does not exist.

NEED	TODAY	WHEN
Hosting	Nothing chosen. Two containers plus Postgres and Redis.	BLOCKER
Managed Postgres	Docker on one laptop. Needs automated backups, point-in-time recovery, and a tested restore drill — the single number that should trigger moving images out of the database is how long a restore takes, and 728 MB of image bytes ride along with every one.	BLOCKER
Secret manager	<code>.env</code> files on a laptop. Needs a real store and a rotation policy before any live key exists.	BLOCKER
TLS and domain	Everything is localhost. The CSP is now derived from <code>NEXT_PUBLIC_MEDUSA_URL</code> rather than hardcoded, so this is configuration rather than code — but it is untested against a real origin.	BLOCKER
CI to staging	CI runs every suite on push and builds both images. It deploys nothing, and there is no staging environment to deploy to.	BEFORE LAUNCH
Rollback	Migrations are forward-only in practice and no rollback has been rehearsed.	BEFORE LAUNCH
Monitoring and on-call	Nothing is watching. No uptime checks, no alerting, no log aggregation. You are the rotation.	BEFORE LAUNCH
CDN	Images are content-addressed and served <code>immutable</code> with a one-year max-age, so they are CDN-ready and no CDN is in front of them. Every image request currently holds a database connection and competes with checkout.	BEFORE LAUNCH

04

Legal and compliance — not engineering

THE FOUR THAT NO CODE CAN CLOSE

Three of these gate EU revenue outright and one gates personalisation, which is the entire commercial argument for owning this stack (§9.6). **They have lead times and can all be started today, in parallel with everything else on this list.** The storefront already refuses EU addresses and names which appointment is missing, so the build is not waiting on them — the revenue is.

REQUIREMENT	CONSEQUENCE WHILE MISSING	WHEN
IOSS registration + EU intermediary	Import VAT on EU B2C consignments cannot be collected correctly. EU sales blocked.	BLOCKER
GDPR Article 27 EU representative	Required to sell into the EU from the US. EU sales blocked.	BLOCKER
GPSR EU Responsible Person	Without one, apparel cannot lawfully be placed on the EU market at all.	BLOCKER
Licensing position	Unresolved, and printing a player's name to order raises it more sharply than selling a stock shirt does. This gates personalisation going live, not being built.	BLOCKER
US economic nexus registrations	Stripe Tax calculates, it does not register. Nexus reaches 45 states.	BEFORE LAUNCH
Legal review of the policies	Privacy, terms, refunds and shipping are published with a visible draft banner. A refund policy is a contract term; the banner is honest, not sufficient.	BEFORE LAUNCH

Trader details still outstanding

The policy pages render each missing field as a labelled gap with the reason it is required, rather than omitting the disclosure. Three remain:

- **Registered postal address** — required on the storefront and on every invoice. The live store publishes none either.
- **Privacy / data-subject request channel** — GDPR Art. 15–22 and the US state laws all require one.
- **A domain mailbox.** Support currently resolves to a personal Gmail address, and the trade name (*Crux Christi*) differs from the brand (*Find Any Jersey*). Both are lawful and neither is a good long-term answer.

Already done, and worth not re-litigating

Consent is opt-in with Global Privacy Control honoured silently and automatically; the US state opt-out page exists with a working control; CSP is written for PCI SAQ A-EP with every entry justified; reviews carry provenance and cannot be presented as verified when they are not; no compare-at price, countdown or stock claim appears anywhere that the data cannot support.

05

Data and supplier inputs

ONE EMAIL CLOSES FIVE GAPS

Fibre composition, country of origin, HS codes, care instructions and chest/length measurements. Four of those are regulatory and one is the highest-leverage line on the storefront under a final-sale policy. **All 4,323 products have every regulatory column empty.**

FIELD	COVERAGE	BLOCKS	WHEN
Fibre composition	0 of 4,323	EU Textile Regulation	BLOCKER
Country of origin	0 of 4,323	Customs declarations	BLOCKER
HS code	0 of 4,323	Cross-border shipping	BLOCKER
EU responsible person	0 of 4,323	GPSR — EU market entry	BLOCKER
Size measurements	none	Size guide, and returns we cannot accept	BEFORE LAUNCH

FIELD	COVERAGE	BLOCKS	WHEN
Print-file format	unknown	Every queued personalisation	BLOCKER

Catalog quality

MEASURE	VALUE	NOTE
Published products	4,323	Synced against the live store on 29 August
Flagged <code>needs_review</code>	157	Mostly unresolvable ambiguity, not error — “Los Angeles Basketball” is the Lakers or the Clippers, and a wrong team is worse than a missing one
No team resolved	113	Excluded from the team facet until reviewed
Exactly one image	67%	A decision (§13.6), with a stated cost
Custom blanks	69	Printing included at \$89.99
Imported reviews	84	Average 4.94, matching the source aggregate exactly
... attached to a product	17	The rest name no product or one we no longer stock. Not a migration failure — there is no more per-product data in existence
In our catalog, not on the live store	1,076	Left alone deliberately. A delisted product is not necessarily unsellable on a sourcing model

06

The order to do it in

Grouped by what unblocks what, not by size. Nothing in the first group takes long, and nothing in the later groups can start without it.

DAY ONE — COSTS NOTHING, UNBLOCKS EVERYTHING

1. **Decide R2.** Build the adapter, or set `criticalInProduction: false`. Until then production cannot start.
2. **Start the four legal appointments.** IOSS, Article 27, GPSR responsible person, and a licensing opinion. These have lead times measured in weeks and cost nothing to begin.
3. **Send the supplier email.** Five gaps, one message.
4. **Get a domain mailbox** and a registered address into the config.

THEN — THE KEYS, IN THIS ORDER

1. **Stripe test keys** and `stripe listen`. Finishes step 8 of the payment path and lets the whole funnel be exercised end to end for the first time.
2. **Resend.** Order confirmations do not exist until this is set.
3. **Shippo** for labels and tracking; **Stripe Tax** switched on and verified against a real account.

THEN — INFRASTRUCTURE, BEFORE ANY LIVE KEY

1. Hosting, managed Postgres, Redis for the event bus and locking.
2. Secret manager. **No live key should ever touch a laptop.**
3. A restore drill that actually restores. Then decide whether 728 MB of image bytes stay in the database, using the restore time as the number.
4. Staging, then CI deploying to it, then a rehearsed rollback.
5. Sentry, uptime checks, log aggregation.

1. Legal review of the four published policy documents.
2. Screen-reader and keyboard pass.
3. A test order through the real Stripe account, with a real label bought and a real refund issued.
4. Decide who staffs the personalisation and returns queues, and what the turnaround target is.
Both exist and neither has an owner.

NOT ON THIS LIST, BECAUSE IT IS FINISHED

Checkout and cart, discount codes, five shipping regions with a real rate card, tax calculation, fulfilment rating, customer accounts with password reset, self-service returns with an admin queue, personalisation with a live preview and a human review gate, 4,323 products with 6,437 images in Postgres, 11 curated collections, SEO with a 4,527-URL sitemap and structured data, consent and privacy controls, eight admin screens, and 537 tests that run in CI on every push.

Three things worth deciding, not just doing

The business case has not changed shape

`research.md` §1 is unambiguous: on cost alone this is a losing trade. At 10,000 orders/month the own stack runs roughly **\$4,725/month more expensive** than Shopify, and the entire justification rests on personalisation revenue — which needs only **\$0.47 of margin per order** to clear that bar. The live store has since validated the price point independently: it sells custom blanks at \$89.99 against a \$64.99 base, a \$25 uplift that brackets the \$19.99 bundle the spec proposed as a guess.

That makes the print-file format and the licensing opinion the two most valuable open items on this entire document, and neither is engineering.

Negotiate Stripe before committing

§1 calls this the single cheapest thing on the list. Interchange-plus pricing above meaningful volume is the only lever that closes the processing gap, and it is worth asking for before the build is sunk rather than after.

The returns policy reversed, and it was the right call

§12.4 recommended free size exchange as the category's answer to a money-back guarantee. The live store publishes *"all sales are final"*, and the code now follows the published policy because a checkout promising one thing and a policy page saying another is worse than either. One constant reverses it. **But under final sale, the missing size measurements stop being a data-quality gap and become the customer's loss** — which is why that supplier email matters more than its position on a list of five fields suggests.